

PAYSON, AZ, USA

WMO: 723745

Lat: 34.257N

Lon: 111.339W

Elev: 5158

StdP: 12.16

Time Zone: -7.00 (W07)

Period: 96-19

WBAN: 00374

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.7	25.4	-8.2	4.3	44.6	-3.6	5.5	47.7	22.3	38.5	18.5	36.1	4.8	30	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	94.6	60.1	91.4	59.8	90.2	59.7	66.8	78.9	65.7	78.5	64.7	78.4	9.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	108.4	68.3	63.0	104.6	68.6	61.0	97.4	69.4	34.5	79.1	33.5	77.7	32.7	77.7	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.6	15.3	12.7	DB	16.0	98.8	5.1	2.2	12.3	100.4	9.3	101.8	6.5	103.0	2.7	104.6
			WB	13.9	68.7	4.7	1.3	10.5	69.6	7.8	70.4	5.2	71.1	1.8	72.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.1	42.5	44.1	50.7	56.5	62.0	74.8	77.3	75.5	70.7	60.8	50.2	42.6
	DBStd	14.08	6.99	8.24	6.43	6.29	6.86	5.15	3.60	3.60	4.77	6.20	7.17	7.24
	HDD50	858	239	195	68	17	5	0	0	0	0	7	84	243
	HDD65	3420	696	585	442	260	136	2	0	0	10	151	443	695
	CDD50	4164	7	30	91	213	377	744	846	790	620	342	91	13
	CDD65	1253	0	0	0	6	43	296	381	325	180	22	0	0
	CDH74	13803	0	0	11	135	739	4039	3808	3143	1659	265	4	0
	CDH80	5796	0	0	0	10	182	1954	1758	1325	532	35	0	0
Wind	WSAvg	4.8	4.4	4.6	5.0	5.9	6.1	5.5	4.7	4.3	4.3	4.5	4.3	3.9
Precipitation	PrecAvg	21.6	2.1	1.9	2.3	1.1	0.6	0.4	2.4	3.1	2.1	1.8	1.9	2.2
	PrecMax	32.7	7.0	9.1	7.5	4.5	1.9	2.9	4.3	7.3	6.7	10.3	5.3	9.4
	PrecMin	14.9	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.7	0.0	0.0	0.0	0.0
	PrecStd	5.0	1.7	1.9	1.9	1.1	0.6	0.6	1.0	1.9	1.7	1.9	1.4	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.0	70.2	75.3	81.3	88.3	99.3	96.8	94.9	90.6	84.0	73.5	64.5
		MCWB	43.7	46.8	49.4	50.5	55.4	59.7	62.4	60.7	60.4	54.9	48.2	43.6
	2%	DB	61.2	66.4	72.0	78.5	84.4	95.3	93.3	91.3	88.3	79.5	70.1	61.5
		MCWB	41.9	45.7	48.2	49.4	53.7	59.0	61.4	61.7	59.3	52.8	47.5	42.5
	5%	DB	57.5	63.2	68.0	74.7	81.5	92.6	91.0	90.1	86.0	76.9	66.3	57.5
		MCWB	40.6	43.8	46.0	48.1	51.6	57.6	61.5	61.8	58.1	52.1	46.4	42.1
	10%	DB	54.7	58.6	64.2	72.0	78.8	90.2	89.7	87.8	82.5	73.3	63.3	54.3
		MCWB	39.6	42.1	44.0	47.2	50.3	56.9	61.3	61.7	58.5	51.0	45.4	40.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.2	50.4	51.1	53.0	57.9	64.5	68.3	67.5	66.7	60.9	51.8	49.9
		MCDB	58.9	64.4	71.0	73.7	85.1	86.2	77.8	79.9	77.3	64.7	64.6	56.0
	2%	WB	44.6	47.0	49.1	51.0	55.4	62.2	67.2	66.2	64.4	58.0	50.0	46.5
		MCDB	53.8	61.6	68.5	74.1	80.8	86.5	78.6	79.3	76.5	68.0	63.8	54.8
	5%	WB	42.8	45.2	47.3	49.3	53.1	60.7	66.3	65.3	63.2	56.1	48.2	44.5
		MCDB	53.0	59.7	63.9	71.2	77.8	85.5	78.4	79.0	76.3	69.2	62.4	53.6
	10%	WB	41.2	43.3	45.5	47.9	51.2	59.1	65.3	64.4	61.9	54.2	46.7	42.7
		MCDB	52.7	54.9	61.9	68.9	75.9	84.4	78.1	78.9	75.7	68.3	60.4	51.6
Mean Daily Temperature Range	5% DB	MDBR	19.5	19.0	22.6	24.0	25.5	27.8	22.9	22.7	22.3	21.8	20.0	16.8
		MCDBR	23.6	25.0	26.4	28.1	30.6	30.2	25.5	26.3	26.1	25.3	23.2	21.3
		MCWBR	12.3	12.6	12.1	12.5	12.0	10.9	8.2	8.0	9.0	11.0	11.3	11.7
	5% WB	MCDBR	18.9	21.5	23.8	26.0	26.7	25.2	20.2	21.3	20.1	19.2	20.2	18.2
		MCWBR	11.2	11.7	11.5	11.7	11.4	9.7	6.7	7.3	8.3	9.6	10.6	10.7
Clear-Sky Solar Irradiance	taub	0.246	0.251	0.273	0.286	0.297	0.302	0.354	0.350	0.304	0.283	0.253	0.246	
	taud	2.680	2.683	2.594	2.545	2.556	2.538	2.494	2.484	2.610	2.619	2.671	2.681	
	Ebn at Noon	316	323	320	318	313	309	293	293	305	306	310	310	
	Edh at Noon	22	24	29	32	32	33	34	34	29	26	22	21	
All-Sky Solar Radiation	RadAvg	1032	1326	1805	2257	2537	2671	2184	2033	1880	1521	1164	926	
	RadStd	94	120	117	96	142	120	132	100	75	115	62	60	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (14 neighbors)	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146	+59	

Nomenclature: See separate page